

Qiushi Engine

Large-Scale Autonomous Discovery of Kissing Number Constructions

Complete Research Report

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Abstract

The kissing-number problem is a classical problem in discrete geometry whose exact solution is known in only a few dimensions. Recent artificial-intelligence approaches have begun to discover improved configurations through large-scale numerical and combinatorial search, but converting such searches into general mathematical constructions and rigorous proofs remains challenging. Here we use Qiushi Engine, an autonomous multi-agent research system, to investigate kissing numbers and obtain new lower bounds in nineteen dimensions: 25, 27, 32–39, 43, 45, and 49–55. The resulting constructions arise from distinct structural mechanisms, including coordinated motions of contact layers, labelled direction reuse, joint support exchanges, signed-code replacements, cross-shell lattice constructions, low-overlap lattice isometries, and spherical-design moment certificates. These methods yield, among others, $K(25) \geq 197580$, $K(27) \geq 201567$, $K(38) \geq 591900$, $K(43) \geq 2553792$, $K(45) \geq 7380090$, and $K(55) \geq 53301140$. The autonomous system carried out the construction searches, mathematical analysis and computational verification, while each final result was reduced to explicit mathematical arguments and independently checkable finite certificates. Our results illustrate how autonomous research systems can move beyond optimization within fixed formulations to discover new mathematical representations and constructions at scale.

Keywords Autonomous mathematical discovery; kissing number; geometric deformation; constant-weight code; lattice lifting; signed design; spherical design

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1 Introduction

The kissing-number problem is one of the oldest problems in discrete geometry. In 1694, Isaac Newton and David Gregory discussed whether 12 or 13 equal spheres could simultaneously touch a central sphere in three dimensions. The answer, 12, was proved more than two centuries later [1]. Yet the apparent simplicity of the question is deceptive. More than three hundred years after its formulation, exact kissing numbers are known only in dimensions 1, 2, 3, 4, 8 and 24. In the exceptional dimensions 8 and 24, the optimal configurations arise from the E_8 and Leech lattices, with 240 and 196560 contacts, respectively [1, 2]. In most other dimensions, substantial gaps remain between the best constructions and known upper bounds.

Progress on lower bounds has historically relied on explicit mathematical structure. Lattices and error-correcting codes provide large families of vectors whose pairwise interactions can be controlled simultaneously [2, 3]. These ideas have led to laminated-lattice constructions, lattice sections and extensions, code-based spherical configurations, and lifting methods that combine a high-dimensional mother configuration with additional coordinates [4–8]. Such constructions can contain hundreds of thousands or even tens of millions of vectors. Their strength, however, is also a source of difficulty: once a configuration is highly constrained, improving it may require reorganizing an entire collection of points rather than inserting a single new one. The relevant search spaces grow combinatorially, while the geometric degrees of freedom that permit improvement can be difficult to identify from the original representation.

Artificial intelligence has recently begun to alter how these spaces are explored. AlphaEvolve, a general-purpose coding agent, discovered a 593-point kissing configuration in dimension 11, improving the previously known lower bound of 592 [9]. PackingStar formulated the kissing-number problem as a two-player matrix-completion game and used game-theoretic reinforcement learning to search the space of Gram matrices, reporting new constructions across dimensions 25–31 and uncovering thousands of candidate geometric structures [10]. EinsteinArena allowed autonomous agents to submit solutions, inspect intermediate results and build on discoveries made by other agents; this collective process increased the dimension-11 lower bound from 593 to 604 [11]. The Station subsequently reported exact 604-point configurations and structural theorems developed in an open-world multi-agent environment [12]. Together, these developments demonstrate the scope of AI-guided geometric search and collaborative mathematical discovery.

These advances also expose a deeper challenge. Finding an improved configuration is only one part of mathematical discovery. Why does the configuration exist, which structural degree of freedom makes it possible, can the mechanism be generalized, and how can the result be converted into a proof or an independently verifiable certificate? Computational searches begin from a prescribed representation—for example, coordinates, Gram matrices, codes or a fixed optimization objective. Open-ended mathematical research additionally requires the ability to change that representation when the current one becomes limiting, to formulate new intermediate problems, connect apparently unrelated structures, and develop arguments that certify the resulting construction.

Autonomous scientific research systems provide a possible route towards this broader form of discovery. Recent large-language-model-based agents can carry out extended sequences of reasoning, computation and tool use, while revising research strategies in response to intermediate evidence. Qiushi Engine was previously used for end-to-end autonomous scientific discovery on a physical optical platform [13], where it conducted long-horizon investigations that combined theoretical reasoning, computation and experiment. Here we ask whether the same paradigm can be extended to open mathematical research: can an autonomous research system move

beyond optimizing a fixed mathematical formulation and instead discover new representations, constructions and proofs?

Here we report an autonomous investigation of the kissing-number problem that yields new lower bounds in nineteen dimensions:

$$25, \quad 27, \quad 32\text{--}39, \quad 43, \quad 45, \quad 49\text{--}55.$$

The resulting bounds include

$$\begin{aligned} K(25) &\geq 197580, & K(27) &\geq 201567, & K(38) &\geq 591900, \\ K(43) &\geq 2553792, & K(45) &\geq 7380090, & K(55) &\geq 53301140. \end{aligned}$$

Rather than arising from a single optimization procedure, these improvements were obtained through a sequence of distinct structural discoveries. They include coordinated motions of complete contact layers, labelled liftings that separate direction deletion from reuse, joint support exchanges that exploit shared deletion costs, signed-code replacements generated through Hadamard transfers, compatible directions drawn from a different lattice shell, enlarged lattice-line classes combined through low-overlap isometric images, and spherical-design moment identities and inequalities that determine or bound large lattice sections without enumerating the ambient shell.

A common pattern connects these apparently different constructions. Configurations that appear saturated in one representation retain exploitable freedom in another. In dimension 25, moving an entire 552-point contact layer while preserving its internal Gram matrix creates room for an additional point. In dimension 27, labelled lifting separates direction reuse from deletion; with the first layer fixed, jointly selecting second-layer directions and labels enlarges the final two-layer construction. In dimensions 32–37 and 39, optimizing collective exchanges and individual sign patterns exposes improvements hidden by separate support-insertion scores and complete sign bundles. In dimension 38, changing the search from the minimal Leech shell to a second lattice shell produces 288 additional points. In dimensions 43 and 45, counting millions of minimal lattice vectors reduces to a finite space of projection signatures whose multiplicities satisfy exact spherical-design moments. A single enlarged line class in dimension 48, combined with low-overlap isometric images, similarly generates new constructions throughout dimensions 49–55.

Qiushi Engine carried out the construction searches, mathematical analysis and computational verification. Each discovered mechanism is expressed through explicit geometric or combinatorial conditions. Each reported bound is supported by finite witnesses checked using exact integer and rational arithmetic, lattice and code identities, rigorous interval arithmetic, and finite combinatorial verification. In dimension 25, outward-rounded interval arithmetic certifies the remaining strict inequalities [14]. These mathematical arguments and reproducible certificates make the constructions independently checkable.

These results suggest a role for AI in mathematics that extends beyond accelerating numerical search. For difficult problems with enormous structured search spaces, a central task may be to discover the representation in which progress becomes possible. Autonomous research systems that can alternate between exploration, structural interpretation and rigorous verification may provide a route towards such discoveries at scale.

1.1 The spherical formulation

The kissing number $K(n)$ is the largest number of unit balls that can touch a central unit ball in $\mathbb{R}^n$ without overlapping interiors. Dividing their centres by two gives the equivalent spherical problem

$$K(n) = \max\{|X| : X \subset S^{n-1}, \langle x, y \rangle \leq \frac{1}{2} \quad (x \neq y)\}.$$

Thus the angle between distinct directions is at least 60° .

1.2 Results

Theorem 1.1. *For every dimension n in Table 1, there is a kissing configuration of N_n points, where N_n is the lower bound in the second column. Thus $K(n) \geq N_n$.*

The constructions in this report give the following lower bounds. The comparison column uses the fixed public sources recorded on 2026-09-24 to 2026-09-27.

Table 1 Explicit lower bounds and their public comparison values.

Dimension	Lower bound	Public comparison	Increase	Source
25	197580	197579	1	[15]
27	201567	201566	1	[16]
32	347584	346944	640	[17]
33	363968	362048	1920	[17]
34	384196	381124	3072	[17]
35	409676	409548	128	[17]
36	484760	484568	192	[17]
37	498024	496232	1792	[17]
38	591900	591612	288	[15]
39	763668	756116	7552	[15]
43	2553792	2545056	8736	[18, 19]
45	7380090	7379838	252	[18, 19]
49	52430156	52430140	16	[15]
50	52458468	52458418	50	[15]
51	52500930	52500816	114	[15]
52	52585772	52585516	256	[15]
53	52698696	52698222	474	[15]
54	52923774	52922906	868	[15]
55	53301140	53299730	1410	[15]

The dimension-32 comparison uses Lysenstøen’s 1671-word code listed in Brouwer’s table [17]. The ERS formula gives $2^{17} + 128 \cdot 1671 + 2 \cdot 32 \cdot 31 = 346944$. The 1667-word code of Echols [20] remains the seed used in our search.

1.3 Construction principles

The nineteen bounds arise from seven connected mechanisms. Each changes a specific mathematical object and has a corresponding compatibility or counting argument.

Coordinated motion in dimension 25. A contact layer of 552 points has fixed transverse coordinates and a common latitude. Moving the latitude together with the extra coordinate

preserves every internal product. The horizontal projection remains on chords joining old shell points, so convexity controls the unchanged equator. Two cap repairs complete the construction and increase the count from 197579 to 197580. The gain comes from moving retained points, not from selecting another point in a fixed pool.

Direction reuse and joint labels in dimension 27. A lifted direction contributes several copies but incurs only one equatorial deletion. Tail geometry determines the permitted reuse. The four-triangle construction realizes the capacity of a prescribed twelve-label model. The larger two-layer construction then combines a fixed first layer with a new joint selection of second-layer owners and labels. Selecting 311 owners instead of 310 gives 201567 points. The capacity statement and the final two-layer count concern different stages of the same lifting geometry.

Shared deletion costs in dimensions 32–34, 37 and 39. For a compatible family Y of new supports, the old supports to remove form a union $F(Y)$. The gain is $|Y| - |F(Y)|$. Several insertions may therefore succeed together even when none improves the count alone. The new support families have sizes 1676, 1803, 1960, 2845 and 3383, respectively. Their intersections and the distances of the accompanying dense codes establish the complete layered constructions.

Signed design transfer in dimensions 35–37. Coordinate regions with controlled interaction with the retained layers admit larger signed replacements. Pairing coordinates and applying a Hadamard transform carries transverse weight-four blocks to weight-eight signed vectors, with norms and inner products scaled by the same factor. The resulting codes have 256, 576 and 512 points. The first attains the eleven-coordinate capacity; the second follows directly from four colours of two one-factorizations. In dimension 37, thirteen extra supports contribute 1664 points and the replacement adds 128 more.

Cross-shell completion in dimension 38. The base lifting uses all minimal Leech directions but leaves an empty equator. For a norm-48 vector v and a norm-32 vector u in the same integer Leech lattice, the minimum norm gives $\|v \pm u\|^2 \geq 32$, hence $|v \cdot u| \leq 24$. This removes the entire old shell from the subset-selection constraints. The remaining task is to find mutually compatible new lines. The 144-line construction contributes 288 points, giving 591900.

Image unions in dimensions 49–55. An enlarged class of 7077 minimal lines in P_{48p} produces further compatible classes by isometries. The relevant size is their union: a repeated line is assigned once. With S union lines and t distinct antipodal tail lines, the construction has $52416000 + 2S + 2t$ points. One lifting principle thus supplies seven bounds, with separately specified image families and an explicit decomposition of the gain from class enlargement and image selection.

Target moments in dimensions 43 and 45. The minimal shell of an extremal 48-dimensional lattice is a spherical 11-design. Its moments constrain all admissible projection fibres. For every extremal even unimodular lattice of rank 48 containing $\sqrt{3}E_8$, a rational identity determines exactly 2553792 minimal vectors perpendicular to the specified D_5 child and each of its Weyl conjugates. In dimension 45 a rational inequality gives

$$f(e_1) + f(e_2) + f(e_3) \geq 7377408 + 9f(-2, -2, 3).$$

The 298 explicit witnesses in the final fibre imply 7380090 projected points. A neighbourhood-difference calculation on paired parent endpoints makes the small fibre searchable. An exact count and a witness-driven lower bound are two different consequences of the moment equations.

1.4 Mathematical context

The Leech lifting framework of Cohn, Jiao, Kumar and Torquato [7, Theorem 7.5], and its subsequent refinements, combine a mother direction with a low-dimensional tail. PackingStar provides the 496-point blocks used in the four-triangle construction [21, 22]. The 25-dimensional deformation starts from Kravatskiy’s 197579-point configuration; the 27-dimensional two-layer input is Lindow’s 201566-point construction [15, 16]. The empty-equator lifting in dimension 38 and the 7069-line comparison class in dimension 48 also come from [15]. The new objects are the coherent motion and repairs, the second-layer allocation, the added shell lines, and the enlarged class with its selected image families.

The code constructions use the layers of Edelman, Rains and Sloane [6], the dense code of Cheng–Sloane [23], and constant-weight codes and tables [17, 20, 24]. Local replacement has a direct precursor in the free-zone method of Takhanov–Yun [25, Section V]. The 51-block weight-four source is the classical Kalbfleisch–Stanton design [26]. Here the new supports, coordinate pairings, signed selections and boundary embeddings produce the larger configurations.

Lattice sections and the antipode construction originate in Conway–Sloane [4, 5]. The public comparisons for 43 and 45 dimensions are the counts of Sun–Wang [18, 19]; the embedded E_8 analysis of Dorofeev, Sun and Wang [27] supplies the parent geometry. Spherical-design theory [28, 29] provides the moment identities. The calculations here select the target statistics and finite witnesses that yield the stated counts.

1.5 Reading the constructions

The report follows the constructions from their governing geometric relations to the explicit finite objects. The final appendices specify coordinates, checking procedures and research development. The accompanying repository contains the bilingual dimension accounts, source data and programs, so that a reader can reconstruct the objects, check the proofs’ finite hypotheses and vary the construction choices.

2 Coordinated shell motion in dimension 25

A coordinated motion of a complete contact layer enlarges the 197579-point configuration supplied in the public [KissingNumbers/dim25-lens-heads](#) package. The construction moves 552 retained equatorial points, adjoins one point, and replaces two cap points. A whole contact layer moves without changing its internal products. The following observation describes the geometric freedom before any particular coordinates are chosen.

Lemma 2.1 (Coordinated chord motion). *Let n be a unit vector, $a > 0$, and $r_i \perp n$ with $\|r_i\|^2 = \rho - a^2$. Put $u_i = r_i - an$ and $w_i = r_i + an$. For $c^2 + d^2 = a^2$, define $W_i = (r_i + cn, d)$. Then $\|W_i\|^2 = \rho$ and $W_i \cdot W_j = w_i \cdot w_j$. If $z \cdot u_i \leq \rho/2$ and $z \cdot w_i \leq \rho/2$, then $(z, 0) \cdot W_i \leq \rho/2$.*

Proof. Orthogonality gives the norm and product identities. The horizontal component is $\frac{a-c}{2a}u_i + \frac{a+c}{2a}w_i$, a convex combination since $|c| \leq a$. Its product with z obeys the same bound as both endpoints. $\square$

Normalization and input. All points have squared norm 4; the kissing condition is an inner product at most 2 between distinct points. Write the input configuration as

$$C_0 = \{(z, 0) : z \in L \setminus D\} \cup \{(x_i, \pm 1) : 0 \leq i < 1016\} \cup \{P_0, N, S\},$$

where L is the 196560-vector minimal shell of the Leech lattice, $|D| = 1016$, $P_0 = (P, 0)$, and $N = (0, 2)$, $S = (0, -2)$. The head indices refer to rows of the source `heads_X.npy`; the mathematical values of its 44 rational heads are preserved as exact rational numbers in `baseline-heads.json`.

Let v be the first complete block's norm-6 lean and put $n = v/\sqrt{6} = -P/2$. An exact integer description of this direction is

$$v = \frac{1}{\sqrt{8}}(-5, -1, -1, -1, -1, 1, -1, 1, -1, -1, -1, 1, 1, -1, -1, -1, 1, 1, -1, 1, -1, -1, 1, -1).$$

The complete block is $U = \{u \in L : \langle u, v \rangle = -3\}$, of size 552. Every $u \in U$ belongs to D . For each such u , the point $w = u + v$ is minimal, lies outside D , and satisfies $\langle u, v \rangle = -3$ and $\langle u, w \rangle = 1$. The corresponding old cap points have horizontal coordinate $u + tv$, with $t = (3 - \sqrt{3})/6$. All these input identities are checked by the shell and owner data.

The modification. For $u \in U$ define

$$r = u + \sqrt{3/2}n = w - \sqrt{3/2}n, \quad a = \frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{4}, \quad b = \frac{\sqrt{6} - 2}{4}.$$

Then $r \perp n$, $\|r\|^2 = 5/2$, and the old cap points over u are $(r - n/\sqrt{2}, \pm 1)$. Replace every $(w, 0)$ by

$$W(r) = (r + an, b),$$

and adjoin

$$Q = (\sqrt{3}n, -1) = (V/4, -1),$$

where $V = \sqrt{8}v$ is the integer vector printed above. In particular, the new point Q is rational and has squared norm $48/16 + 1 = 4$. Finally replace the upper point $(x_{984}, 1)$ and the lower point $(x_{1008}, -1)$ by R_+ and R_- , respectively. The certificate specifies two integer vectors and defines

$$R_+ = \frac{2k}{\sqrt{1007176}}, \quad R_- = \frac{2\ell}{\sqrt{10209}},$$

where the following integer vectors have squared norms 1007176 and 10209:

$$\begin{aligned} k &= (-336, -9, -298, 286, -17, 15, -39, 22, 277, 19, 364, 304, 9, \\ &\quad -2, -24, -39, 4, 304, -9, -4, 271, -29, 9, -2, 500), \\ \ell &= (-25, -31, -33, 4, 3, 2, 36, 33, 3, -1, -2, -3, -4, \\ &\quad -1, 4, -2, 1, 2, 4, -31, -1, -28, -3, -28, -50). \end{aligned}$$

The opposite old cap points $(x_{984}, -1)$ and $(x_{1008}, 1)$ are retained. These integer vectors specify the repairs exactly.

Lemma 2.2 (Norms and pairs not involving other cap points). *All the $W(r)$ and Q have squared norm 4. They are mutually compatible, and are compatible with all unchanged equatorial points,*

both poles, and P_0 . Each $W(r)$ is also compatible with the old cap points belonging to the complete first block.

Proof. The identities

$$a^2 + b^2 = \frac{3}{2}, \quad \sqrt{3}a - b = 2, \quad -\frac{a}{\sqrt{2}} + b = -\frac{3}{4}$$

give the norm assertions and $\langle W(r), Q \rangle = 2$. For two different block points,

$$\langle W(r_i), W(r_j) \rangle = \langle r_i, r_j \rangle + \frac{3}{2} = \langle w_i, w_j \rangle \leq 2.$$

Since $0 < a < \sqrt{3/2}$, the horizontal component of $W(r)$ is

$$r + an = (1 - \lambda)w + \lambda u, \quad \lambda = \frac{\sqrt{3/2} - a}{2\sqrt{3/2}} \in (0, 1).$$

An unchanged equatorial point z is different from both u and w . Both $\langle u, z \rangle$ and $\langle w, z \rangle$ are at most 2, so their convex combination is at most 2.

The shell has integral $\langle v, z \rangle \in \{-3, -2, -1, 0, 1, 2, 3\}$. The negative extreme layer is deleted and the positive extreme layer is exactly the moved set. Thus every unchanged equatorial point obeys

$$\langle Q, (z, 0) \rangle = \frac{\langle v, z \rangle}{\sqrt{2}} \leq \sqrt{2} < 2.$$

The finite shell data verify these layer identities in integers.

The remaining pure-point products are

$$\langle W(r), P_0 \rangle = -2a, \quad \langle W(r), N \rangle = 2b, \quad \langle W(r), S \rangle = -2b,$$

$$\langle Q, P_0 \rangle = -2\sqrt{3}, \quad \langle Q, N \rangle = -2, \quad \langle Q, S \rangle = 2.$$

All satisfy the required inequality. For the same block index, the products with the old upper and lower cap points are, respectively,

$$\frac{5}{2} - \frac{a}{\sqrt{2}} + b = \frac{7}{4}, \quad \frac{5}{2} - \frac{a}{\sqrt{2}} - b = \frac{11}{4} - \frac{\sqrt{6}}{2} < 2.$$

For different block indices, $\langle r_i, r_j \rangle \leq 1/2$, so both products are smaller still. □

The finite certificate. The previous lemma handles every equality created by the modification. The remaining products to certify are:

1. the 552 moved points against the 2030 unchanged cap points;
2. Q against those unchanged cap points;
3. R_+ and R_- against every point retained in the final configuration, including one another, with their own diagonal entries omitted.

Outward-rounded rational upper bounds for these four groups are

$$1.999372, \quad 1.965926, \quad 1.994722, \quad 1.987522.$$

Each is below 2. The checker uses 192-bit outward-rounded Arb intervals and integer comparisons, and checks both normalizations exactly. Pairs involving only unchanged points retain the inequalities of the input configuration.

Theorem 2.3. *The specified configuration consists of 197580 distinct points on the radius-2 sphere in $\mathbb{R}^{25}$ with pairwise inner products at most 2. Consequently the kissing number in dimension 25 is at least 197580.*

Proof. The lemma, the verified finite checks, and the independently checked source certificate exhaust all pairs. The baseline checks include every head-shell and same-cap head-head inequality, with integer or rational acceptance decisions. Compatibility within the Leech shell follows from its lattice minimum. All squared norms are exactly 4, including those of R_+ and R_- by definition. The 552 moved points replace 552 existing points and the two repairs replace two existing cap points; only Q increases the inventory. Thus

$$|C| = 197579 - 552 - 2 + 552 + 2 + 1 = 197580.$$

No duplicate can occur: two equal norm-4 vectors would have inner product 4, contradicting the verified bound 2 for their distinct indices. $\square$

3 Labelled lifts: count the directions that remain

Let $C \subset S^{D-1}$ be a kissing configuration. Split the larger space as $\mathbb{R}^D \oplus \mathbb{R}^k$. Choose pairwise distinct nonzero tails t_i of norm less than one and subsets $A_i \subseteq C$. Write $a_i = \sqrt{1 - \|t_i\|^2}$ and $U = \bigcup_i A_i$. Retain $(x, 0)$ for $x \in C \setminus U$, insert $(a_i x, t_i)$ for $x \in A_i$, and add a pure-tail kissing code $Y \subset S^{k-1}$.

The count is

$$|X| = |C| - |U| + \sum_i |A_i| + |Y|.$$

Every original direction is deleted once, however many labels use it. For distinct labelled pairs $(x, i) \neq (x', j)$ and pure tails $y \in Y$, the remaining conditions are

$$a_i a_j \langle x, x' \rangle + \langle t_i, t_j \rangle \leq \frac{1}{2}, \quad \langle t_i, y \rangle \leq \frac{1}{2}.$$

These conditions separate tail geometry from head selection without discarding their coupling.

3.1 Multiplicity and the four-triangle construction

If every tail has squared norm $h^2 < 1/2$, repeated copies of one direction require

$$\langle t_i, t_j \rangle \leq h^2 - \frac{1}{2}.$$

For m copies, the squared norm of the sum of their tails is at most

$$\frac{m}{2}(1 - m(1 - 2h^2)).$$

Its nonnegativity implies $m \leq \lfloor (1 - 2h^2)^{-1} \rfloor$. At $h^2 = 1/3$, a direction can be used three times. Equality is realized by a zero-sum triangle.

The twelve roots of D_3 , scaled to tail norm $1/\sqrt{3}$, split into four zero-sum triangles. Four disjoint 496-point Leech blocks, each carried on one triangle, give

$$196560 - 4(496) + 12(496) + 12 = 200540.$$

The comparison allocation used multiplicities $(3, 3, 2, 2, 2)$. The new allocation uses $(3, 3, 3, 3)$, preserving 496 additional equatorial points while keeping the same number of lifted points. The blocks themselves are published PackingStar data; the change is their assignment to the tail geometry.

For this fixed twelve-label system with at most α directions per label, the count is at most $196560 + 8\alpha + 12$. Thus the construction attains the allocation capacity at $\alpha = 496$.

3.2 The later 201,567-point construction

The later construction extends a public two-layer configuration. Its first layer, taken unchanged from Lindow's 201,566-point contribution, has 2,258 integer heads of squared norm 192. Of these, 2,090 delete one Leech minimal vector each, and 168 delete none. Each head produces three caps.

Use the integer Leech convention in which minimal vectors have squared norm 32 and the kissing threshold is 16. A first-layer head y produces

$$(y/3, 4v/\sqrt{3}),$$

where v runs through its assigned D_3 triangle. The exact head thresholds are 48 on one triangle and 96 between different triangles.

A second-layer owner u is a Leech minimal vector assigned to one of three coordinate lines. Replace its equatorial point by

$$(\sqrt{3}u/2, \pm\sqrt{8}e_j).$$

The supplied 311-owner witness has line sizes 105, 103, and 103. Every first-layer head satisfies $y \cdot u \leq 32$. Two owners on one line have product at most eight; different lines have product at most sixteen. Owners are distinct and do not overlap the first-layer deletion set.

For example, the first/second-layer product is at most

$$\frac{\sqrt{3}}{6} 32 + \frac{4\sqrt{8}}{\sqrt{3}} = \frac{16\sqrt{3} + 8\sqrt{6}}{3} < 16.$$

The second-layer/pure-tail product is at most $\sqrt{8}\sqrt{32} = 16$, including equality. The two caps of one owner also have product sixteen. These checks give

$$196560 - 2090 - 311 + 3(2258) + 2(311) + 12 = 201567.$$

Replacing the public input's 310 second-layer owners by the new selection of 311 adds one point: each owner replaces one equatorial point by two caps.

4 Layered codes, joint exchanges and signed constructions

These seven results use one geometric construction, with improvements at different levels. Dimensions 32, 33 and 34 use larger compatible families of eight-element supports; dimension 39 combines such a family with an explicit sign code. Dimensions 35, 36 and 37 also change the signs carried by the supports. The important transition is from deciding whether to retain a whole bundle to replacing it by a different finite set.

4.1 Compatibility of the layers

The Edel–Rains–Sloane construction [6] places three coordinate patterns on the same unit sphere. Let $n \geq m \geq 32$, let $T \subset \{0, 1\}^m$ be a binary code, and let $\mathcal{B}$ be a family of eight-element subsets of $\{0, \dots, n-1\}$. A support records the nonzero coordinates of a vector; the signs specify the vector itself. The dense, middle and root layers are

$$\begin{aligned} X_T &= \left\{ \frac{1}{\sqrt{m}}((-1)^{c_0}, \dots, (-1)^{c_{m-1}}, 0, \dots, 0) : c \in T \right\}, \\ X_{\mathcal{B}} &= \left\{ \frac{1}{\sqrt{8}} \sum_{i \in B} \varepsilon_i e_i : B \in \mathcal{B}, \varepsilon_i \in \{-1, 1\}, \prod_{i \in B} \varepsilon_i = 1 \right\}, \\ X_R &= \left\{ \frac{\pm e_i \pm e_j}{\sqrt{2}} : 0 \leq i < j < n \right\}. \end{aligned}$$

Each support carries $2^7 = 128$ even-parity sign patterns. The layers have respectively m , eight and two nonzero coordinates, so they are disjoint. Their union contains

$$N = |T| + 128|\mathcal{B}| + 2n(n-1) \quad (4.1)$$

points.

Proposition 4.1 (Three-layer construction [6]). *If T has minimum Hamming distance at least $\lceil m/4 \rceil$ and distinct supports satisfy $|B \cap B'| \leq 4$, then the union is a kissing configuration in dimension n .*

Proof. Two dense-layer points have product $1 - 2d_H(c, c')/m \leq 1/2$. Distinct even-parity patterns on the same support differ in at least two positions, giving at most $1 - 4/8 = 1/2$. Different supports give at most $|B \cap B'|/8 \leq 1/2$. Distinct roots either share at most one coordinate with the same sign, or have the same support and differ in at least one sign; their product is also at most $1/2$.

There are three remaining cross-layer cases. Dense–middle products are at most $8/(\sqrt{m}\sqrt{8}) = \sqrt{8/m} \leq 1/2$; dense–root products are at most $2/(\sqrt{m}\sqrt{2}) = \sqrt{2/m} \leq 1/2$; middle–root products are at most $2/(\sqrt{8}\sqrt{2}) = 1/2$. These cover all distinct pairs. $\square$

The geometric problem reduces to finding a large binary code and a large support family with intersection at most four. The support indicators have constant weight eight and satisfy $d_H(1_B, 1_{B'}) = 16 - 2|B \cap B'|$. Thus the intersection condition is minimum distance eight in the constant-weight code. Equation (4.1) converts improvements of such codes into kissing-number lower bounds [17, 20, 24].

4.2 Joint exchanges of supports

Inserting a new support generally requires removing old supports with large intersection. For a single candidate, these deletions may consume the gain. Several mutually compatible candidates can instead be obstructed by the same old supports. The search therefore selects additions jointly.

For a candidate b , put $F(b) = \{B \in \mathcal{B} : |B \cap b| > 4\}$. If Y is disjoint from the old family and its internal intersections are at most four, then

$$\begin{aligned} \mathcal{B}' &= (\mathcal{B} \setminus F(Y)) \cup Y, & F(Y) &= \bigcup_{b \in Y} F(b), \\ |\mathcal{B}'| - |\mathcal{B}| &= |Y| - |F(Y)|. \end{aligned} \tag{4.2}$$

The cost is a union size, not the sum of the individual conflict counts. The stored final exchange in dimension 33 removes nine supports and adds ten; in dimension 34 it removes six and adds seven; an exchange in dimension 37 removes 37 and adds 40. The data contain the parent families and actual deletions and additions, so these unions can be reconstructed.

There is also a direct discrete formulation. For a finite candidate set $\mathcal{C}$, let $z_b \in \{0, 1\}$ select a candidate and let $u_B \in \{0, 1\}$ delete an old support. Impose $z_b + z_{b'} \leq 1$ for incompatible candidates and $z_b \leq u_B$ for $B \in F(b)$, and maximize

$$\sum_{b \in \mathcal{C}} z_b - \sum_{B \in \mathcal{B}} u_B.$$

An optimal choice deletes only supports actually blocking selected candidates, so its objective is exactly the gain in (4.2). This identifies the useful search property: concentrate conflicts on shared old supports, rather than minimizing each candidate's conflict count separately. Candidate generation and joint selection are complementary tasks; changing the candidate family changes the constructions available to this finite optimization problem.

The blocking sets also organize candidate generation. For a chosen integer f , consider eight-subsets outside $\mathcal{B}$ with $|F(b)| \leq f$. If the pool includes all such subsets, it covers every positive exchange with at most $f + 1$ additions: for $b \in Y$, $|F(b)| \leq |F(Y)| < |Y| \leq f + 1$. The threshold therefore describes a specific exchange neighbourhood rather than a limit on the final code.

Candidate conflicts admit a compact representation. Two eight-subsets intersect in at least five positions exactly when they share a five-subset. Each candidate occupies $\binom{8}{5} = 56$ such subsets, and the constraints are

$$\sum_{b \in \mathcal{C}: Q \subset b} z_b \leq 1 \quad \left(Q \in \binom{\{0, \dots, n-1\}}{5} \right).$$

They may be added successively: solve the current selection problem, inspect the selected five-subsets, and add the constraints at repeated subsets. Once none repeats, the selection is compatible and a positive objective gives an exchange. The recorded dimension-34 pool contains 10715 candidates with at most three blockers; this procedure yields the seven-for-six exchange above. After each accepted exchange, recomputing the blocking sets makes further replacements available.

The following parameters give direct three-layer constructions. The dimension-37 support family was enlarged from 2843 to 2845 and then modified by a signed replacement, whose final count is given below.

Table 2 Parameters of the direct three-layer constructions.

n	Dense length m	$ T $	$ \mathcal{B} $	Points
32	32	131072	1676	347584
33	32	131072	1803	363968
34	32	131072	1960	384196
39	39	327680	3383	763668

Each extra support contributes 128 spherical points. For example, $131072 + 128 \cdot 1676 + 1984 = 347584$ in dimension 32. The gains of 15 and 24 supports in dimensions 33 and 34 give respectively 1920 and 3072 points over the comparison constructions.

The length-32 layer uses the Cheng–Sloane [32, 17, 8] code [23], whose generator expands to $2^{17} = 131072$ words. Dimension 39 uses $T = \bigcup_{a=1}^{10} (H + r_a)$, where H has dimension 15 and minimum weight 12. Enumeration of the supplied kernel and representatives gives

$$\min_{h \in H} \text{wt}(h + r_a + r_b) = 10$$

for all 45 distinct coset pairs. The cosets are disjoint, with $10 \cdot 2^{15} = 327680$ words and minimum distance ten, as required. The kernel is obtained by shortening an extended quadratic-residue code; the programs compare generator row spaces and check the actual distances.

The listed dimension-39 comparison has the same dense-layer size. Increasing its 3324 supports to 3383 adds $128(3383 - 3324) = 7552$ points. Recording the support and dense layers separately identifies the finite objects and their respective contributions.

4.3 Hadamard transfer and local replacement

The support count is not the only variable. Each original support carries all 128 even-parity patterns. Removing a few complete bundles and inserting a different signed set can fill the affected region more efficiently. Both the internal products and products against the retained configuration must be controlled.

Proposition 4.2 (Paired Hadamard transfer). *Let $\mathcal{A}$ be a family of four-element supports with intersections at most two, transverse to a fixed coordinate pairing. Applying a Hadamard matrix to each pair in all signed versions of $\mathcal{A}$ gives $16|\mathcal{A}|$ weight-eight vectors of squared norm eight and distinct-pair products at most four.*

Proof. Start with four-element supports having pairwise intersections at most two, and attach all sixteen sign patterns to each. Pair the coordinates and retain supports meeting each pair in at most one coordinate; call them transverse. Apply

$$H = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

to each coordinate pair. Both $(\varepsilon, 0)$ and $(0, \varepsilon)$ become two nonzero coordinates, so weight four becomes weight eight. Distinct input vectors have product at most two: different supports satisfy the intersection condition, and distinct sign patterns on the same support differ in at least one position. Since $H^T H = 2I$, the images have squared norm eight and pairwise products at most four. Invertibility preserves their number. $\square$

After division by $\sqrt{8}$, these vectors form an internally compatible spherical set Q . Intersections at most four with each retained support ensure compatibility with the retained middle layer. The dense and root cross bounds remain $\sqrt{8/32} = 1/2$ and $1/2$, independently of the parity of the new signs. This is why replacements need not consist of complete even-parity bundles.

The free-zone replacement principle appears in Takhanov–Yun [25, Section V, Definition 5 and Lemma 4] for signed weight-four codes. Here it is applied to weight-eight regions inside the ERS middle layer. The choices of region, transverse supports and paired transform give the replacements below. In dimensions 35 and 36, boundary compatibility can be secured before choosing Q . Take a coordinate region W , remove the old supports contained in it, and require $|B \cap W| \leq 4$ for every retained support. Any new weight-eight vector on W is then compatible with the retained middle layer. Only its local internal products remain to be controlled. The regions are $W = \{12, \dots, 22\}$ and $W = \{12, \dots, 23\}$, respectively, with zero-based coordinates. They contain one and three old supports. The deletion costs are fixed at 128 and 384 points, turning a problem involving the whole configuration into a smaller independent construction problem.

Direct search among signed weight-eight vectors in eleven and twelve coordinates has respectively $2^8 \binom{11}{8} = 42240$ and $2^8 \binom{12}{8} = 126720$ candidates. Weight-four transfer replaces this search with the choice of a combinatorial source and a coordinate pairing, while preserving an algebraic proof of the inner-product bounds. Dimension 35 starts with a Steiner quadruple system $S(3, 4, 10)$: each triple belongs to exactly one block, giving $\binom{10}{3}/4 = 30$ blocks with pairwise intersections at most two. Enumerating the 945 perfect matchings on ten coordinates yields sixteen transverse blocks. For dimension 36 we use the 51-block construction of Kalbfleisch–Stanton [26, Theorem 7], also described in [25, Section V, Construction 1]. Split twelve coordinates into two sets of six and use the same five one-factors of K_6 in each half. For every colour, the nine choices of a left and a right edge give a four-element block, producing 45 crossing blocks. Add, in each half, the complements of the three edges of the first colour to obtain six further blocks.

The resulting 51 blocks still have intersections at most two. Same-colour crossing blocks sharing one edge intersect in two points; different colours share at most one endpoint in each half. Internal blocks in the same half intersect in two points; those in different halves are disjoint. An internal block meets a crossing block in at most two points.

Proposition 4.3 (A 576-point local code). *The paired Hadamard image of 36 transverse blocks from the two one-factorizations is a 576-point weight-eight code in twelve coordinates.*

Proof. Choose the three edges of one colour as the coordinate pairs in each half. Each edge of the other four colours meets every chosen pair at most once. Its union with a same-colour edge from the other half is therefore transverse. There are $4 \cdot 3 \cdot 3 = 36$ such blocks; their intersections are at most two by the preceding argument. Proposition 4.2 gives $36 \cdot 16 = 576$ points. The six internal blocks are unnecessary for this transverse subfamily. $\square$

In the supplied local coordinate order, the chosen pairs are

$$(0, 1), (2, 4), (3, 5), (6, 7), (8, 10), (9, 11).$$

They form the same colour in the two halves. The direct construction with these pairs regenerates exactly the stored 576 vectors. The accompanying program also enumerates all 945 and 10395 matchings of the two source codes, recovering the 256-point and 576-point sets. Thus the

existence proof is explicit, while the exhaustive calculation records the finite choices available in this model.

The three constructions select 16, 36 and 32 transverse supports, producing 256, 576 and 512 new points. The compatible region in dimension 35 has eleven coordinates; ten form five pairs and the remaining coordinate is zero. Dimensions 36 and 37 use twelve coordinates each. The explicit pairs and transverse supports regenerate all 256, 576 and 512 signed points for comparison with the final sets.

Table 3 Signed replacements. The support count precedes the deletion of bundles.

Dimension	Original supports	Bundles removed	Points added	Net gain
35	2157	1	256	128
36	2742	3	576	192
37	2845	3	512	128

Combining the support choices and replacements gives

$$\begin{aligned}
K(35) &\geq 131072 + 128(2157 - 1) + 256 + 2(35)(34) = 409676, \\
K(36) &\geq 131072 + 128(2742 - 3) + 576 + 2(36)(35) = 484760, \\
K(37) &\geq 131072 + 128(2845 - 3) + 512 + 2(37)(36) = 498024.
\end{aligned}$$

Relative to the comparison with 2832 supports, dimension 37 gains 1664 points from thirteen extra supports and 128 from the replacement, totalling 1792. Dimensions 35 and 36 keep their original support counts; their gains come entirely from the signed choices.

The local choice in dimension 35 also attains an exact capacity bound. Let $M(11)$ be the largest number of weight-eight vectors in $\{0, \pm 1\}^{11}$ with distinct-pair products at most four. Fix a full sign string $\sigma \in \{\pm 1\}^{11}$. Two vectors agreeing with σ on their supports would intersect in at least five positions and have product at least five. Thus each sign string extends at most one selected vector. Each vector, in turn, has eight extensions on its three zero coordinates. Counting gives

$$8|Q| \leq 2^{11}, \quad M(11) \leq 256.$$

The 256-point construction attains equality, so $M(11) = 256$. This explains the complete use of the eleven-coordinate local region. In twelve coordinates, two eight-element supports can intersect in only four positions, allowing a sign string to extend several vectors; this is the geometric opening used by the larger local code in dimension 36.

The finite checks use the coordinate order, pairs and four-subsets specified in the data. In dimension 37, a support string numbered from one at the left has coordinate i at integer-mask bit $37-i$. The program regenerates all 512 points from 32 transverse supports and six coordinate pairs, checks equality with the stored replacement, and checks intersections with every retained support. The origin, count and geometric conditions are thereby connected through the same input.

These seven results treat the support family and its sign patterns as coupled choices. The next construction changes another part of the geometry: equatorial directions left available by a large lattice lifting.

5 Equatorial completion of a Leech-lattice lifting

The signed replacements balance additions against deletions. Here a large lifting leaves an unused equator. We seek directions that can be inserted directly, compatible both with one another and with the entire lifted configuration. A set of 144 such lines contributes 288 points, increasing the lower bound from 591612 to 591900.

5.1 The lifted configuration and its equatorial constraints

Let $\Sigma \subset \mathbb{Z}^{24}$ be the squared-norm-32 minimal Leech shell, with 196560 vectors. The supplied extended Golay code generates 1104 vectors with two entries ± 4 ; 97152 vectors with entries ± 2 on a weight-eight support and even sign parity; and 98304 vectors obtained from signs $(-1)^{c_i}$ by reversing one chosen sign and multiplying that entry by three. The Golay generator has rank 12 and weight enumerator $1 + 759z^8 + 2576z^{12} + 759z^{16} + z^{24}$. Distinct vectors in this standard shell model have inner product at most 16 [2].

Write $\mathbb{R}^{38} = \mathbb{R}^{24} \oplus \mathbb{R}^{14}$. The public 591,612-point construction uses every minimal Leech direction in a cap layer [15], but leaves the equatorial subspace available for other directions. The new construction exploits this distinction.

Partition the 98,280 antipodal lines of Σ into 644 classes whose representatives satisfy $|u \cdot u'| \leq 8$ within each class. In $\mathbb{R}^{14}$ take 1932 vectors of squared norm 8 and distinct inner products at most 4, partitioned into 644 zero-sum triangles. Each triangle has pairwise products -4 . Pair a head class with a triangle and form six points per head line:

$$\left(\frac{\varepsilon u}{\sqrt{48}}, \frac{d}{\sqrt{24}} \right), \quad \varepsilon \in \{1, -1\},$$

with d ranging over the assigned triangle. These are unit vectors. Equal heads with different tails give product $32/48 - 4/24 = 1/2$; different heads with an equal tail give at most $8/48 + 8/24 = 1/2$. Points from different classes give at most $16/48 + 4/24 = 1/2$, and the other cases give smaller bounds. The cap layer has $6 \cdot 98280 = 589680$ points.

Both signs on a head line are included. Opposite heads contribute $-32/48$, so even with identical tails their total product is only $-1/3$.

Add a pure tail layer $(0, Rd/\sqrt{8})$, using the same 1932 directions and an exact rational rotation $R = Q/D$. Its integer data satisfy

$$QQ^\top = D^2 I, \quad |d \cdot Qd'| \leq \sqrt{48} D.$$

The first identity preserves the pure tail products; the second bounds cap–tail products by $1/2$. Integer square roots suffice to check the latter exactly. This gives the base count $589680 + 1932 = 591612$.

5.2 Compatible lines in the second shell

We use the norm-48 shell of the same Leech lattice. If $\mathcal{G}$ is the supplied Golay code, its integer coordinate model is

$$\mathcal{L} = \left\{ x \in \mathbb{Z}^{24} : \begin{array}{l} x_i \equiv m \pmod{2} \text{ for some } m \in \{0, 1\}, \\ ((x_i - m)/2 \bmod 2)_{i=1}^{24} \in \mathcal{G}, \\ \sum_i x_i \equiv 4m \pmod{8} \end{array} \right\}. \quad (5.1)$$

Linearity of $\mathcal{G}$ and its all-one word make this set a lattice. Its minimum squared norm is 32. Indeed, for even coordinates a nonzero Golay residue has at least eight entries of absolute value two. A zero residue makes all coordinates multiples of four, and the sum congruence excludes a single entry of absolute value four. For odd coordinates, a vector of squared norm below 32 would have every entry ± 1 ; the doubly even Golay weight then makes its coordinate sum zero modulo eight, contrary to the last congruence. The minimal vectors described above attain 32.

Lemma 5.1 (Compatibility of the second shell). *If $v \in \mathcal{L}$ has squared norm 48 and $u \in \mathcal{L}$ has squared norm 32, then $|v \cdot u| \leq 24$.*

Proof. The distinct norms imply $v \pm u \neq 0$. Their squared norms are $80 \pm 2v \cdot u \geq 32$, which gives the claimed bound. $\square$

Although all minimal lines are used, equatorial points can come from directions of another length. Squared norm 48 gives the same normalization denominator as the cap heads. The new compatibility conditions consequently become integer-product checks with threshold 24. The 144 supplied vectors satisfy all three congruences in (5.1). It remains to check their mutual products:

$$\|v_i\|^2 = 48, \quad |v_i \cdot v_j| \leq 24 \quad (i \neq j), \quad |v_i \cdot u| \leq 24 \quad (u \in \Sigma).$$

Proposition 5.2 (Equatorial completion). *Adding the 288 points $(\pm v_i/\sqrt{48}, 0)$ to the cap and pure-tail layers gives a 591900-point kissing configuration in dimension 38.*

Proof. The 288 new equatorial points $(\pm v_i/\sqrt{48}, 0)$ are internally compatible. Their products with a cap are $\pm v_i \cdot u/48 \leq 1/2$, and they are orthogonal to the pure tails. No base point lies on the equator, so no deletion is needed. Thus

$$K(38) \geq 591612 + 2 \cdot 144 = 591900.$$

$\square$

Lemma 5.1 removes the entire old shell from the search constraints. Generate integer norm-48 vectors satisfying the lattice congruences and retain one canonical representative per antipodal line. Join two representatives when $|v \cdot v'| > 24$. The remaining problem is an independent-set problem in this finite graph: each selected line contributes exactly two equatorial points. A joint replacement deletes the union of the old neighbours of the inserted lines, so it can improve the set even when no single line can be added. Every accepted set is checked for mutual compatibility. The verifier also recomputes all products with Σ , independently checking the cross-shell lemma on the supplied data; these products need not be recalculated during subset selection.

There is a direct geometric explanation. Write $C = \Sigma/\sqrt{32}$ for the unit mother shell. An equatorial direction $z = v/\sqrt{48}$ must satisfy $|z \cdot c| \leq \sqrt{3/8}$. This threshold is strictly larger than $1/2$, because the head of a mixed point has length $\sqrt{2/3}$. The space available at the equator of the lifting is therefore larger than the space for inserting unit points directly into the mother shell. The second-shell directions exploit this space. Using every minimal direction need not exhaust the possible points in the resulting 38-dimensional configuration.

The search first produced 142 compatible lines and then 144. The completed data use the same Golay basis as the cap construction. Canonical line representatives have positive first nonzero coordinate and are sorted lexicographically. This convention binds the class labels, tail triangles and added-line checks to one coordinate system.

6 Seven dimensions from one enlarged line class

The catalogue lattice P_{48p} is even unimodular of dimension 48 and minimum squared norm six. Its theta series determines a minimal shell of 52,416,000 vectors [30, 31].

6.1 Disjoint blocks from overlapping images

The new finite object is a class of 7,077 minimal lines. Integer representatives c in the catalogue Gram basis satisfy $cGc^t = 6$ and

$$|cGc'^t| \leq 2 \quad (c \neq \pm c').$$

Exact lattice isometries give further classes. Select a family of images and assign every line to its first occurrence. This produces disjoint line subsets with union size S . Taking both signs of every line produces compatible head blocks.

Use the root codes $A_1, A_2, D_3, D_4, D_5, E_6, E_7$ as tails in dimensions one through seven. Their sizes are 2, 6, 12, 24, 40, 72, 126. Assign each head block to a different antipodal tail line. A used mother line loses its two equatorial points and contributes four cap points at head squared norm $3/4$ and tail squared norm $1/4$. The count is

$$K(48 + k) \geq 52416000 - 2S + 4S + \tau_k = 52416000 + 2S + \tau_k.$$

k	Selected images	Union size S	Tail points τ_k
1	1	7,077	2
2	3	21,231	6
3	6	42,459	12
4	12	84,874	24
5	20	141,328	40
6	36	253,851	72
7	63	442,507	126

6.2 Why the lift preserves all angular constraints

Let x be a selected mother vector divided by $\sqrt{6}$, and let p_i represent its assigned unit tail line. The four points replacing $\{\pm x\}$ are

$$\left(\varepsilon \frac{\sqrt{3}}{2} x, \delta \frac{1}{2} p_i \right), \quad \varepsilon, \delta \in \{\pm 1\}.$$

They have unit norm. Within a block, distinct head lines have absolute product at most $1/3$; in the entire minimal shell the corresponding bound is $1/2$. These two facts give all cap inequalities:

$$\begin{aligned} \text{different lines in one block:} & \quad (3/4)(1/3) + 1/4 = 1/2, \\ \text{same head, opposite tails:} & \quad 3/4 - 1/4 = 1/2, \\ \text{opposite heads:} & \quad -3/4 + 1/4 = -1/2, \\ \text{different blocks:} & \quad (3/4)(1/2) + (1/4)(1/2) = 1/2. \end{aligned}$$

The last line uses distinct tail lines. Reusing a tail line for arbitrary different blocks would permit $5/8$. A cap meets a retained equatorial point with product at most $\sqrt{3}/4$, and a pure tail with product at most $1/2$. The retained equator and pure tails already satisfy the kissing condition. Thus the construction and its count are established by the class inequalities and the lattice minimum.

6.3 Separating the two sources of gain

The gain over the public 7,069-line class [15] has two sources: eight additional mother lines and improved choices of isometric images. Apply the selected images to the original class and call the union size S_0 ; let S_{ref} be the comparison construction's union size. Then

$$N - N_{\text{ref}} = 2(S - S_0) + 2(S_0 - S_{\text{ref}}).$$

The first term holds the images fixed and measures class growth. The second holds the original class fixed and measures image selection.

Dimension	Class growth	Image selection	Total increase
49	16	0	16
50	48	2	50
51	96	18	114
52	192	64	256
53	320	154	474
54	572	296	868
55	996	414	1410

The three images in dimension 50 are disjoint. In dimension 55, overlap reduces the $8 \cdot 63$ possible new lines to 498 distinct lines; image selection on the original class supplies another 207. This is why enlarging one class and maximizing the union are complementary parts of the construction.

7 Extremal lattice sections and design moments

The preceding chapters specify each layer and check its products. Here we select vectors from a large known lattice shell and obtain lower-dimensional configurations by projection. The required information is the number of vectors with particular projection signatures. Dimension 43 counts shell vectors perpendicular to a five-dimensional section; dimension 45 projects three fibres of a three-dimensional section. Both become finite multiplicity problems.

7.1 Mother shells, projection domains and moments

An extremal even unimodular lattice in dimension 48 has minimum squared norm six. Its theta series is

$$E_4^6 - 1440E_4^3\Delta + 125280\Delta^2 = 1 + 52416000q^3 + O(q^4).$$

Thus its minimal shell has 52416000 vectors. Here $E_4 = 1 + 240q + 2160q^2 + 6720q^3 + \dots$ and $\Delta = q - 24q^2 + 252q^3 + \dots$; the constant term and vanishing first two coefficients determine

the expression in weight 24 [2]. Venkov's theorem makes this shell a spherical 11-design [29, Theorem 4.1].

A spherical 11-design has the same averages as the sphere of equal radius for every polynomial of degree at most eleven. Thus its low-degree moments along any chosen direction are known without listing its vectors. For example,

$$\sum_{x \in X} (x \cdot t)^2 = \frac{6|X|}{48} \|t\|^2, \quad \sum_{x \in X} (x \cdot t)^4 = \frac{3 \cdot 6^2 |X|}{48 \cdot 50} \|t\|^4.$$

Grouping vectors by projection turns the left sides into linear combinations of unknown multiplicities, with completely known right sides.

For section vectors t_i with Gram matrix H , a shell vector has signature $y_i = x \cdot t_i$ and projection norm $y^\top H^{-1} y \leq 6$. The signatures are integral when the section vectors lie in the lattice. Inner-product bounds against known short section vectors further restrict this finite domain. For degree $2m \leq 10$,

$$\sum_{x \in X} \prod_{\ell=1}^{2m} (x \cdot t_{i_\ell}) = \frac{|X| 6^m}{48 \cdot 50 \cdots (48 + 2m - 2)} \sum_{\mathcal{P}} \prod_{(a,b) \in \mathcal{P}} H_{i_a i_b},$$

where $\mathcal{P}$ ranges over pairings of the labelled positions; degree zero gives the shell size. Unknown signature multiplicities therefore satisfy rational linear equations. In dimension 43 these equations determine the target statistic exactly. In dimension 45 they are combined with non-negativity to bound the statistic from below.

7.2 Dimension 43: counting a child section through its parent

Let X be the minimal shell of P_{48n} [32]. A supplied integer matrix embeds $\sqrt{3}E_8$ into this lattice. Its first five simple roots span a $\sqrt{3}D_5$ section U , and $X \cap U^\perp$ lies in dimension 43.

The root numbering is fixed by the following standard coordinates:

$$\alpha_1 = \frac{1}{2}(1, -1, -1, -1, -1, -1, -1, 1), \quad \alpha_2 = e_1 + e_2, \quad \alpha_j = e_{j-1} - e_{j-2} \quad (3 \leq j \leq 8).$$

The supplied 8×48 integer matrix T and ambient Gram matrix G satisfy $(TGT^\top)_{ij} = 3\alpha_i \cdot \alpha_j$; rows 1, 2, 3, 4, 5 specify the child. This fixes the embedding as well as the abstract D_5 root type. The use of an embedded $\sqrt{3}E_8$ parent is developed in Dorofeev–Sun–Wang [27]; here its projection data are used to count this D_5 complement.

Proposition 7.1 (E_8 -extension count). *Let L be any extremal even unimodular lattice of rank 48 containing an isometric copy of $\sqrt{3}E_8$, and let X_L be its squared-norm-six shell. Let U_L be the span of the first five embedded simple roots in the numbering above. Then*

$$|X_L \cap U_L^\perp| = 2553792.$$

The same count holds for every $W(E_8)$ -conjugate of this D_5 subsystem inside the embedded E_8 . The supplied matrices G, T realize the hypotheses in P_{48n} , giving $K(43) \geq 2553792$.

No direction change is needed. Distinct minimal vectors satisfy $\|x - y\|^2 \geq 6$, so $x \cdot y \leq 3$; division by $\sqrt{6}$ gives a kissing configuration. The task is to count $|X \cap U^\perp|$ exactly.

For a minimal vector, its projection onto the parent section is $\lambda/\sqrt{3}$ with $\lambda \in E_8$. The norm bound and the inner products with every embedded root give a complete domain of 26,641 possible signatures. Of these, 240 are exceptional signatures of the section roots themselves.

This domain exhausts the possible projections. Nonexceptional λ satisfy $\|\lambda\|^2 \leq 18$ and $|\lambda \cdot r| \leq 3$ for every E_8 root r . Enumeration gives shells of squared norm 0, 2, 4, 6 and 8, of sizes 1, 240, 2160, 6720 and 17280 respectively. Adding the 240 exceptions $\lambda = 3r$ gives the complete domain.

The extremal 48-dimensional shell is an 11-design. For projection coordinates, every even moment through degree ten is therefore the corresponding spherical moment. Averaging over the parent Weyl group yields multiplicities

$$1092000, \quad 116235, \quad 9180, \quad 495, \quad 63/4$$

on its five nonexceptional norm orbits. As orbit averages of integer fibre sizes, these multiplicities can be rational.

The perpendicular child has 1, 12, 6, 24, 0 signatures on those orbits and twelve exceptional roots. Hence the average child count is

$$1092000 + 12(116235) + 6(9180) + 24(495) + 12 = 2553792.$$

Thus at least one child has at least 2,553,792 points. To count the specified child exactly, average only over a subgroup preserving that child. Explicitly, use the group generated by $-I$, the reflections in $\alpha_1, \dots, \alpha_5$, and the reflections in the twelve E_8 roots perpendicular to all five. The latter roots form A_3 ; each reflection is $s_r(x) = x - (x \cdot r)r$. These formulas determine the group without reference to an ambient automorphism. The complete domain has 43 orbits under this group. Rational moment rows form a matrix A , with right side b and target c . The certificate gives w satisfying

$$A^t w = c, \quad w^t b = 2553792.$$

This proves Proposition 7.1: the signature domain and moment equations depend only on the embedded $\sqrt{3}E_8$, integrality, the lattice minimum and the spherical-design moments. The subgroup need not act by automorphisms of the whole mother lattice: it preserves the signature domain, the moment equations, and the target statistic.

The calculation determines a linear statistic rather than every fibre. Even if $Av = b$ has several solutions, $c^T v$ is constant on them whenever c^T lies in the row space of A . Independent moment rows can first be selected by modular linear algebra, followed by a rational solution for the multiplier. Reproduction then requires just the two displayed rational identities. The complete signature domain remains essential: the moments on the right describe the whole shell, while orthogonality belongs in the target vector c , not in a preliminary deletion of all other fibre columns.

The comparison section has the same abstract D_5 Gram type. What changes is the embedding, which can change the perpendicular population. Its extension to an E_8 parent supplies additional projection constraints that certify this population. The additional equations determine a count of the fixed set; they do not change that set.

7.3 Dimension 45: 298 witnesses control the projected count

Use a specified quadratic-residue even neighbour of rank 48. The code-distance and parity argument in Appendix A proves its minimum squared norm six. Its shell again has 52,416,000 vectors and is an 11-design.

Three section vectors have Gram matrix

$$H = 8I_3 - 2J_3, \quad H^{-1} = (I_3 + J_3)/8.$$

Let $f(y)$ count shell vectors of signature y . The nonexceptional domain is explicitly

$$D_0 = \{y \in \mathbb{Z}^3 : |y_i| \leq 3, \quad |y_1 + y_2 + y_3| \leq 3, \quad \sum_i y_i^2 + (\sum_i y_i)^2 \leq 48\}. \quad (7.1)$$

The section roots are $\pm t_1, \pm t_2, \pm t_3$ and $\pm(t_1 + t_2 + t_3)$. Inner products with these eight vectors give the linear bounds in (7.1), while $y^\top H^{-1} y \leq 6$ gives its quadratic bound. Their own signatures form $E = \{\pm H e_1, \pm H e_2, \pm H e_3, \pm H(1, 1, 1)^\top\}$. Each has multiplicity one: its section projection already has squared norm six. Enumeration of the displayed three-coordinate domain gives $|D_0| = 231$ and $|D_0 \cup E| = 239$.

For each pair $\{y, -y\} \subset D_0$, take the lexicographically smaller representative and set $v_{[y]} = f(y) - f(-y)$; retain zero separately. There are $(231 + 1)/2 = 116$ variables. For a multi-index a of even total degree at most ten, its moment row is

$$A_{a,[y]} = (2 - \mathbf{1}_{y=0}) \prod_i y_i^{a_i}, \quad b_a = M_a - \sum_{y \in E} \prod_i y_i^{a_i},$$

where M_a is the spherical moment defined above. There are $\sum_{j=0}^5 \binom{2j+2}{2} = 161$ such monomials. This constructs all entries of the 161×116 system $Av = b$.

The fibre count $f(y)$ counts shell vectors satisfying $(x \cdot t_1, x \cdot t_2, x \cdot t_3) = y$. The target is $f(e_1) + f(e_2) + f(e_3)$, not every individual multiplicity. After moving known exceptional contributions to the right, write the moment equations as $Av = b$, with $v \geq 0$. For a statistic $c^\top v$, a rational vector w with $c - A^\top w \geq 0$ gives

$$c^\top v = w^\top b + (c - A^\top w)^\top v \geq w^\top b.$$

This is the form of the moment certificate. Choosing the statistic well makes an accessible fibre appear with positive coefficient in the lower bound, so a small explicit witness set becomes useful.

Proposition 7.2 (A positive fibre contribution). *The signature multiplicities satisfy*

$$f(e_1) + f(e_2) + f(e_3) \geq 7377408 + 9f(-2, -2, 3).$$

Proof. Give c coefficient +1 on the three classes $[e_i]$, coefficient -9 on $[(-2, -2, 3)]$, and zero elsewhere. Order the representatives and multi-indices lexicographically. The multiplier in `dual.json`, supplied in Appendix B, then has 161 rational entries and satisfies, by exact matrix multiplication, $c - A^\top w \geq 0$ and $w^\top b = 7377408$. The preceding identity with $v \geq 0$ gives the assertion. $\square$

Corollary 7.3 (The 45-dimensional construction). *The specified quadratic-residue neighbour and section give $K(45) \geq 7380090$.*

Proof. The certificate supplies 298 distinct integer lattice vectors of squared norm six and signature $(-2, -2, 3)$. Thus the right side is at least 7,380,090.

Project the three fibres e_1, e_2, e_3 onto the section's orthogonal complement. The projection formula is $\pi(x) = x - \text{proj}_U x$, giving

$$\pi(x) \cdot \pi(x') = x \cdot x' - y^\top H^{-1} y'.$$

For $y = e_i$, the squared section projection norm is $1/4$; for distinct e_i, e_j , their section projections have product $1/8$. Since distinct mother-shell vectors have product at most three, each projected squared norm is $23/4$. Products are at most $11/4$ within a fibre and $23/8$ between different fibres. Dividing by $\sqrt{23/4}$ gives a kissing configuration:

$$K(45) \geq 7377408 + 9(298) = 7380090.$$

□

A witness search in a Pless lattice gave 292 vectors and the bound 7,380,036. Enlarging the choice of anchors led to 298 vectors in a quadratic-residue even neighbour, with each additional witness contributing nine points to the projected bound. To organize the search, fix a norm-eight anchor $p \in \Lambda$ and set

$$A_p = \{x \in X : x \cdot p = 4\}.$$

This parent set has exactly 2256 points, with completeness certified by low-degree moments. Since $x \pm p$ are nonzero lattice vectors, minimum six gives $|x \cdot p| \leq 4$. The polynomial

$$P(t) = t^2(t^2 - 1)(t^2 - 4)(t^2 - 9)$$

vanishes at all allowed integral products except ± 4 , where it equals 20160. The eighth spherical moment gives $\sum_{x \in X} P(x \cdot p) = 90961920$. Antipodal symmetry makes the two endpoint fibres equal, hence $|A_p| = 90961920/(2 \cdot 20160) = 2256$. The involution $x \mapsto p - x$ preserves A_p . The graph vertices are its 1,128 pairs $\{x, p - x\}$ centred at $p/2$. Two pairs are adjacent when representatives have inner product one or three; exchanging an endpoint preserves this condition. Fibre searches become common-neighbour calculations in this graph.

Use the integer numerator coordinates $p = y/\sqrt{12}$ of the appendix. The selected y has six coordinates of absolute value three and 42 of absolute value one, so $y^2 = 96$. This odd-coordinate shape explores another part of the neighbour rather than repeating anchors from its integral sector. The resulting graph has degree 368 and a selected nonadjacent pair with 70 common neighbours. The difference $368 - 70 = 298$ identifies the witness fibre. The correspondence is explicit. Denote the selected endpoints by r, s and use the section $T = (-s, s - p, p - r)$. For a pair adjacent to the vertex of r but not that of s , choose its endpoint a with $r \cdot a = 3$; then $s \cdot a = 2$. Set $w = p - a$. It satisfies

$$\|w\|^2 = 6, \quad (-s \cdot w, (s - p) \cdot w, (p - r) \cdot w) = (-2, -2, 3).$$

Different pairs give different witnesses, so the 298 selected graph vertices yield precisely the required lattice vectors. The parent-graph checker in Appendix B reconstructs the section and

exactly the 298 vectors in the final certificate. The rational moment inequality then gives the stated bound.

8 Construction principles and further questions

The nineteen bounds arise from several distinct ways of enlarging a constrained configuration. Tight contacts do not always prevent motion: the 552-point layer in dimension 25 moves with its internal Gram matrix unchanged. A saturated selection need not exhaust its candidate geometry: recolouring changes the second-layer selection in dimension 27, while shared blockers make a family of support insertions profitable. A local capacity bound can be attained by importing a different combinatorial object, as the weight-four designs and their Hadamard images show.

Two further mechanisms use information supplied by the ambient lattice. Its minimum makes cross-shell compatibility automatic in dimension 38. Its design moments determine a section count in dimension 43 and give a witness inequality in dimension 45. In the latter case, the right statistic changes the scale of the search: 298 explicit vectors certify a configuration with more than seven million points. For dimensions 49–55, a common class supplies seven lifts, and exact union counts distinguish the gain from class growth from the gain from image selection.

These results suggest structural questions as well as larger finite searches. Which contacting layers admit a boundary-compatible motion? When can a signed code attaining its local capacity be embedded in a larger configuration? Which moment functionals isolate a count that is accessible through a small witness set? Each question retains the geometric constraints responsible for the results while opening choices beyond the specific coordinates presented here.

Qiushi Engine developed these constructions through sustained autonomous mathematical research, connecting geometric freedom, discrete selection and exact counting. The resulting proofs and finite objects make those connections available for further mathematics: they can be reconstructed, varied and used as starting points for new constructions.

A The minimum of the ambient lattice used in dimension 45

Let C be the extended ternary quadratic-residue $[48, 24, 15]$ code [31], and put $M = \{z \in \mathbb{Z}^{48} : z \bmod 3 \in C\}$ and $L_0 = \{z/\sqrt{3} : z \in M, z^2 \equiv 0 \pmod{6}\}$. The supplied basis determines a characteristic numerator $\chi \in M$ with odd coordinates, $\chi^2 = 264$, and $\chi \cdot z \equiv z^2 \pmod{6}$. Define

$$\Lambda = L_0 \cup (\chi/(2\sqrt{3}) + L_0).$$

The characteristic congruence and $\chi^2/12 = 22$ give integrality and evenness; the two index-two constructions give covolume one. The program reconstructs an integer numerator basis B and verifies $BB^\top = 12G$ for the exact Gram matrix used by the 298-vector certificate.

In L_0 , squared norm two or four would require $z^2 \leq 12$. The code minimum weight 15 forces $z \bmod 3 = 0$, whose positive even lattice norms are at least six. In the other coset, $y = \chi + 2z$ has 48 odd coordinates, so $y^2/12 \geq 4$. Equality requires $y_i = \pm 1$.

Let $s_i \in \{\pm 1\}$ represent $\chi_i \bmod 3$. The finite checks give $s \in C$, $\chi \cdot s = 66$, and $\chi_i s_i \equiv 1 \pmod{6}$. Normalizing coordinates by s gives a code containing the all-one word. Its two-symbol enumerator is $1 + 94t^{24} + t^{48}$ [33, equation (20)], so binary-valued codewords have weights 0, 24 or 48. Writing $y = s(1 - 2b)$ gives such a binary word of weight w , but

$$\chi \cdot z = \frac{\chi \cdot s - \chi^2}{2} - \sum_i \chi_i s_i b_i \equiv -99 - w \equiv 3 \pmod{6},$$

contrary to $z/\sqrt{3} \in L_0$. Evenness excludes all other positive norms below six, while $\sqrt{3}(e_1 + e_2)$ has squared norm six. The theta-series and design-moment arguments therefore apply to this specified lattice.

B Construction data and exact verification

This PDF contains the attachment `certificates.zip`. Save it from the PDF viewer's attachment panel, or extract it with `pdfdetach -saveall main.pdf`. It contains the complete finite inputs and the programs used to check them. The English and Chinese editions embed identical archives.

Each construction combines a geometric argument covering all pair types with finite input satisfying its hypotheses. For the code constructions, the input consists of supports, generators, coset representatives and signed vectors. For the lattice constructions, it consists of integer directions, Gram matrices, embedded sections and rational moment certificates. The catalogue `constructions/catalog/results.json` specifies the defining files, exact count and verification tasks for each dimension. The following paths are relative to `constructions/`.

For dimension 25, `d25/` contains the exact baseline heads, the two integer repair vectors, the complete moving-layer proof and the integer, rational and Arb verifiers. The interval checks require `python-flint`. For dimension 27, `d27/data/` gives both head layers and their labels; the independent checker regenerates the shell and verifies all deletion and product conditions. The candidate-graph reconstruction is included. For dimensions 49–55, `p48/data/` supplies the catalogue basis, generators, enlarged class and seven image families. The exact verifier rebuilds every image union and checks tail ranks and the lifting inequalities. The search program also verifies the eight-line augmentation and decomposes the union gain relative to the original class.

For dimensions 32–37 and 39, `codes/data/` contains the complete support lists, binary generators, coset representatives and exchange records. The local replacements for dimensions 35 and 36 are in `codes/hadamard/data/`; their generator file and the reconstruction program reproduce the stored signed vectors exactly. Dimension 37 has its own 512-point replacement and coordinate map in `codes/data/`.

For dimension 38, `d38/base-data/` specifies the Golay generator, Leech line classes, tail triangles and relative rotation defining the 591,612-point base. The 144 added representatives are in `d38/added-lines.txt`. The checker reconstructs the shell, verifies the base, tests the Leech congruences of every new vector, and checks all new mutual and boundary products in exact arithmetic.

For dimension 43, `sections/d43/data/parent-section.txt` gives the ambient Gram matrix and the eight rows embedding the numbered E_8 roots. `sections/d43/data/certificate.json` contains the orbit data, rational moment matrix and multiplier determining the D_5 count. One program reconstructs this system from the geometry; another checks the rational matrix identity independently.

For dimension 45, `sections/qr/` reconstructs the extremal neighbour. Within `sections/d45/data/`, `record.json` contains the section and 298 witnesses, `dual.json` the exact multiplier of Proposition 7.2, and `parent-vectors.json` the 2,256-point parent set. The checkers regenerate

the moment domain and equations, verify lattice membership and inner products, and recover the same witnesses from the parent graph.

After extracting the archive, its `README.md` gives the dependencies and commands. Use a Python environment providing SageMath, NumPy, SymPy and `python-flint`, together with a C++ compiler. In that environment, `python3 tools/reproduce.py --suite all --output ../verification` runs every listed check in a separate directory, without network access. The output records each check and its inputs. The archive's manifest identifies every distributed file; the full repository also contains the report sources and the recorded verification results.

C Construction methods and research development

The mathematical arguments above arise from choices of finite objects, coordinate models and counting methods. This account connects those choices to the candidate families, searches, intermediate constructions and exact calculations that produced the results.

C.1 Moving the retained shell

The 25-dimensional investigation changes the geometry of retained points, rather than restricting every improvement to the original head pool. A complete block of deleted Leech vectors has retained partners on the opposite latitude. Moving all partners together preserves their internal products. Their horizontal coordinates stay on segments between pairs of original vectors, so convexity settles compatibility with the unchanged equator. Two exceptional caps are then replaced by normalized integer vectors. The analytic motion and finite repairs have separate, short verification arguments.

C.2 From assigning blocks to reusing directions

The first Leech lifts retained a large fixed mother shell and used published 496-point compatible blocks. The useful variable was the assignment of these blocks to tail labels. In dimension 25, the latitude at height one half permitted two poles. The count became 197,058 without enlarging the block.

In dimension 27, twelve labels could be partitioned into four zero-sum triangles. Replacing block multiplicities $(3, 3, 2, 2, 2)$ by $(3, 3, 3, 3)$ preserved one more 496-point block at the equator, giving 200,540. The capacity argument then explained the count: a direction could occur at most three times, and equality required full labels with multiplicity three.

The later 201,567-point result uses a different public input. Lindow's contribution had jointly optimized the first and second layers to 201,566. The new witness keeps that first layer and increases the second layer from 310 to 311 owners. The first layer's 2,090 deleting heads and 168 nondeleting heads remain unchanged.

New heads may contribute two or three points after their own deletion costs, and a second-layer owner contributes one. Optimizing only a layer's gross population can select a worse complete configuration.

C.3 From block libraries to image unions above dimension 48

The initial family used 63 disjoint blocks of a Pless neighbour. Its prefix sizes, combined with root tail codes, produced seven configurations in dimensions 49 through 55. The paper retains the neighbour's construction and minimum-norm proof.

The larger family uses the published catalogue basis of P_{48p} . A public class of 7,069 minimal lines was extended to 7,077. Its automorphism images were then selected jointly, using the size of their union as the objective. Assigning repeated lines to their first occurrence produced disjoint subsets without losing the within-class product bound.

The seven union sizes are

$$7077, 21231, 42459, 84874, 141328, 253851, 442507.$$

Each contributes two points per line, in addition to the mother shell and the root tail code. The dimension-50 witness uses three disjoint images. Larger image families retain some overlaps, whose exact effect is recorded in the union sizes.

Different assigned classes use distinct antipodal tail lines. This gives the cross-class product bound $(3/4)(1/2) + (1/4)(1/2) = 1/2$. An injective tail assignment therefore accompanies the union of the selected head classes.

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